

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 3: Atmospheric Density, Temperature & Inversions

Syllabus Source: CAE Oxford (pp. 69-98) | Pass Target: $\geq 90\%$ AviationExam

1. AIR DENSITY & AIRCRAFT PERFORMANCE

Air density (ρ) is mass per unit volume. Lift, drag, engine thrust, and propeller efficiency are directly proportional to air density:

- Density Relationships: Density INCREASES with: Increasing pressure, Decreasing temperature, and Decreasing humidity (dry air is denser than moist air because H₂O molecular weight 18 is lighter than N₂ 28 / O₂ 32!).
- Density Altitude (DA): Pressure altitude corrected for non-standard temperature: Density Alt = Pressure Alt + 120 ft x (OAT - ISA Temp). High density altitude severely degrades take-off roll and climb!

2. TEMPERATURE INVERSIONS (NEGATIVE LAPSE RATE)

An inversion occurs when temperature INCREASES with altitude instead of decreasing:

- Radiation Inversion (Ground Inversion): Forms on clear, calm nights as the ground radiates heat into space. Chills the contact air layer. Disperses mid-morning with solar heating.
- Subsidence Inversion: Forms in high-pressure anticyclones as sinking air warms adiabatically, capping pollution, haze, and stratocumulus clouds below it.
- Frontal Inversion: Occurs across a warm front where warm air ascends over a wedge of cold surface air.
- Aviation Hazards of Inversions: Creates intense low-level windshear across the inversion boundary, severely degrades engine performance climbing through the warm layer, and traps pollutants reducing visibility.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Quick Density Altitude Estimation

Problem: Airport elevation = 4,000 ft. QNH = 1013.25 hPa (Pressure Altitude = 4,000 ft). OAT = +32 C. Step 1: Calculate ISA at 4,000 ft: ISA = 15 C - (4 x 2 C) = +7 C. Step 2: Calculate Delta ISA: 32 C - 7 C = +25 C. Step 3: Apply 120 ft per degree of temperature deviation: 25 x 120 ft = 3,000 ft. Step 4: Density Altitude = Pressure Altitude + Correction = 4,000 ft + 3,000 ft = 7,000 ft! Operational Impact: The aircraft performs as if taking off from an airfield at 7,000 ft!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Density Altitude from QNH and OAT

SCENARIO (Classic Performance & Meteorology Exam Question): An airfield has an elevation of 3,000 ft MSL: - Current QNH = 1003 hPa - Outside Air Temperature (OAT) = +31°C QUESTION: What is the Pressure Altitude and the Density Altitude at the airfield? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate Pressure Altitude (PA): - Formula: $PA = \text{Elevation} + 27 \text{ ft} \times (1013.25 - QNH)$ - Pressure difference = $1013.25 - 1003 = +10.25 \text{ hPa}$. - Altitude correction = $10.25 \text{ hPa} \times 27 \text{ ft/hPa} = 277 \text{ ft}$ (or ~300 ft). - $PA = 3,000 \text{ ft} + 277 \text{ ft} = 3,277 \text{ ft}$ (~3,280 ft). Step 2: Calculate ISA Temperature at 3,277 ft: - ISA Temp at Sea Level = +15.0°C. - Lapse = $3.28 \times 2^\circ\text{C} = 6.56^\circ\text{C}$. - ISA Temp at airfield = $+15.0^\circ\text{C} - 6.56^\circ\text{C} = +8.44^\circ\text{C}$ (~ +8.5°C). Step 3: Calculate ISA Deviation: - Actual OAT = +31.0°C - Delta ISA = $31.0^\circ\text{C} - 8.44^\circ\text{C} = +22.56^\circ\text{C}$ (22.6°C WARMER than standard!). Step 4: Calculate Density Altitude (DA): - Formula: $DA = \text{Pressure Altitude} + 120 \text{ ft} \times \text{Delta ISA}$ - Temperature adjustment = $120 \text{ ft} \times 22.56^\circ\text{C} = 2,707 \text{ ft}$! - $DA = 3,277 \text{ ft} + 2,707 \text{ ft} = 5,984 \text{ ft}$ (~6,000 ft)! FINAL ANSWER: Pressure Altitude = 3,280 ft. Density Altitude = 5,984 ft! (The aircraft will perform as if it were taking off at 6,000 ft altitude!).

[!] EXAM TRAP (AVIATIONEXAM) - Humid Air vs Dry Air Density Trap

- Many students think humid air is heavier than dry air. THIS IS A TRAP! - Water vapor (H₂O, molecular mass 18) is LIGHTER than dry air (N₂ 28 + O₂ 32, average mass 29)! - Therefore, HIGH HUMIDITY DECREASES AIR DENSITY and degrades aircraft performance!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under which weather conditions is a SURFACE RADIATION INVERSION most likely to develop?**

- [A]: Strong winds and heavy overcast clouds
- [B]: CLEAR NIGHTS with calm or very light wind
- [C]: Active cold front passage
- [D]: Midday summer sun

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Rapid terrestrial radiative cooling into a clear sky combined with calm air allows the surface air to chill rapidly while the air above remains warmer.

Q2: How does an increase in moisture content (relative humidity) affect the density of the air?

- [A]: Increases density
- [B]: DECREASES density (moist air is less dense than dry air)
- [C]: Has zero effect
- [D]: Doubles air pressure

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Water molecules (H₂O, mol wt 18) displace heavier nitrogen (28) and oxygen (32) molecules in a given volume, making humid air lighter/less dense.

Q3: What flight hazard is most frequently encountered when climbing or descending through a strong temperature inversion?

- [A]: Severe airframe icing
- [B]: Marked LOW-LEVEL WINDSHEAR and turbulence across the inversion boundary

- [C]: Complete loss of pitot pressure
- [D]: Heavy hail

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The thermal boundary decouples surface air from the upper flow, frequently generating severe vertical and directional windshear.